

Machine Learning Canvas v0.4 — JusticeCast

Goal [Business Understanding]: Quantify how much of bench-reading signal lives in lexical features vs semantic representations on SCOTUS oral arguments.

Decisions

[Business Understanding]

- Per-Justice vote forecasts inform:
 - Pre-argument prep targeting (litigators, week before)
- Amicus brief targeting (post-argument, hours after)
- Historical bench-reading benchmarks (research, continuous)

ML Task

[Modeling]

- Binary stance classification per (case, Justice) row
- Label: voted_petitioner $\in \{0, 1\}$
- Comparative study: BoW vs pre-trained embeddings

Value Propositions

[Business Understanding]

- Empirical lower bound on text-only bench-reading
- Pre-trained semantics beat BoW by $\sim 1.4\text{--}2.1$ pp (10 $\times$ 5 CV, $p < 0.001$)
- Don't sell TF-IDF — sell semantic representations as the floor

Data Sources

[Data Understanding]

- SCDB Justice-Centered file (release 2025_01, Latin-1, 83,644 vote rows $\times$ 61 cols) — labels
- Oyez REST API (2-step fetch: case meta $\rightarrow$ transcript audio JSON, ≤ 1 req/sec) — text
- Both free, public, no auth

Offline Evaluation

[Model Evaluation]

- Test ROC AUC on canonical fold-0
- 5-fold CV mean $\pm$ std (StratifiedGroupKFold by case_id)
- Per-Justice contested-cases AUC (the strict test)
- Per-Justice lift over individual baseline

Making Predictions

[Modeling]

- Per-Justice per-case forecast within hours of oral argument
- Threshold tunable per consumer use case
- Litigators: FN-tilted (recall on petitioner side)
- Amicus: FP-tilted (precision on petitioner side)

Collecting Data

[Data Understanding]

- Per-term refresh after each SCOTUS term ends
- SCDB: single annual download
- Oyez: incremental fetch of new (term, docket) pairs
- Both layers cached on disk (377 MB for the 2005-2024 window)

Features

[Data Preparation]

- Track 1 (BoW): TF-IDF + n-grams (200K-dim sparse) + 424-term custom stopword list + advocate-name regex
- Track 2 (Embeddings): all-MiniLM-L6-v2 (384-dim dense), pre-trained, no fine-tuning, no stopword list
- Identical fold-0 test rows for both tracks (Non-Neg #15)

Live Evaluation & Monitoring

[Model Deployment]

- Track per-term ROC AUC and per-Justice contested AUC
- Flag if AUC drops > 5 pp term-over-term (audit bench composition + case-mix shifts)
- Justice composition changes require justice_id_map updates

Building Models

[Modeling]

- Annual retrain after each SCOTUS term ends (June)
- Re-encode new term transcripts via compute_embeddings.py (~ 12 min CPU)
- Refit LogReg on full historical corpus
- Re-verify SCDB codebook semantics each major release

Stakeholders

[Business Understanding]

- Appellate litigators (pre-argument prep)
- Amicus brief authors (post-argument filing decisions)
- Legal-tech vendors (Lex Machina, Bloomberg Law, Westlaw Edge, SCOTUSblog) — embedded research workflow
- Litigation press (same-day forecast articles)

Headline Result

[Model Evaluation]

- 10 $\times$ 5 repeated CV: $+1.76$ pp $\pm$ 1.31 pp lift ($p < 0.001$)
- 95% CI for mean lift: $[+1.4$ pp, $+2.1$ pp]
- Embeddings win 47 of 50 fold-realizations
- Fold-0 $+3.7$ pp is the lucky end (DeLong $p = 0.023$)
- Effect is small but reliable, not single-fold luck

Each box tagged with its corresponding CRISP-DM phase. Left two columns = Predict (decisions, ML task, eval, predictions, monitoring). Right two columns = Learn (data sources, features, build cadence). Headline result corner = Model Evaluation phase.